

SPRINGER'S INTRODUCTIONS:
INTIMATE CONNECTIONS Why can we do so many things that a mouse cannot? Our brains are not only bigger, but also surprisingly wired. The Holy Grail of neuroscience is the complete mapping of the human brain - the faithful depiction of the network of nerve cells with the tree-shaped branches of the dendrites sprouting from them and the much longer axons, which often serve to transmit signals from a sensory organ or to a muscle fiber. The overall picture includes the tiny nodules on the dendrites; That's where the synapses are located. These are contact and switching points, lively connections to other neurons. Sorting through this thicket down to the level of individual cells and representing it spatially is a gigantic task that until recently seemed utopian. Recently, the young research branch of connectomics (from English: connect) has been able to understand the interaction of neurons better and better.

But is our head just a mouse brain that has been expanded to a thousand times the number of neurons? Or is human nervous tissue perhaps wired differently? To answer this question, the MPI group undertook a detailed comparison of mice, macaques and humans (Science 377, abo0924, 2022). This time, human tissue did not come from epileptics, but from two patients operated on for brain tumors. The researchers wanted to avoid years of treatment with antiepileptic drugs clouding the picture of the synaptic connections. They compared the samples with those of a macaque and five mice.

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H On the one hand, apart from the very obvious quantitative differences such as brain size and number of neurons, there were quite good agreements, which justify the use of animal models. But the MPI team experienced a real surprise on one point.

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Text explaining table above or whatever, anything that does not belong to either of the page columns. This is achieved using three-dimensional electron microscopy. Thin-slice images of cerebral tissue samples can be used to create three-dimensional reconstructions of entire cell groups. Certain nerve cells, the so-called interneurons, are characterized by the fact that they only interact with other nerve cells.